

Delphine VINCENT, PhD

Artificial Intelligence • Machine Learning • Data Science • Bioinformatics

Melbourne, Australia | Australian & French Citizen



Website



Google Scholar



ORCID



LinkedIn



GitHub



Tableau Public



email

SNAPSHOT

- **AI & Data Scientist** with over three years' consulting experience delivering bespoke machine learning, generative AI and analytical solutions for academic and industry collaborators.
- **More than two decades of experience** transforming complex, high-dimensional data into actionable insights through statistics, predictive modelling, experimental design and scientific problem solving.
- **Designs and develops end-to-end AI solutions**, including LLM-powered applications, Retrieval-Augmented Generation (RAG), intelligent chatbots, AI-enabled websites and modern analytical workflows.
- **Experienced Python developer** specialising in machine learning, NLP, workflow automation, API integration and scientific software development.
- **Honorary Senior Fellow**, Faculty of Science, University of Melbourne.
- **Continues professional development through AWS Educate and CodeCademy**, applying cloud technologies to AI deployment, web applications and scalable analytical workflows.
- **71 publications, 13 public datasets, 37 GitHub repositories and 7 Tableau dashboards** demonstrating reproducible, transparent and explainable analytical workflows.
- **Advocate of evidence-based and responsible AI**, combining rigorous validation, reproducibility and explainable analytics to develop trustworthy decision-support solutions.
- **Experienced project leader, scientific editor, mentor and collaborative consultant** who enjoys translating complex technical problems into practical solutions for diverse stakeholders.

EMPLOYMENT HISTORY

Independent AI, Data Science & Bioinformatics Consultant

2023-present | [Independent Research Consultant](#), Melbourne, Australia

- **Partner with researchers, business stakeholders and organisational leaders** to translate complex technical challenges into practical, evidence-based solutions supporting strategic decision making.
- **Deliver bespoke AI, machine learning and data science solutions across diverse scientific and commercial domains**, transforming complex scientific and business questions into reproducible analytical workflows and decision-support tools.
- Design and deliver end-to-end analytical projects spanning **data acquisition, feature engineering, statistical modelling, machine learning, validation, visualisation and communication of actionable insights**.
- **Develop reproducible Python and R applications** for predictive modelling, natural language processing, workflow automation, API integration and scientific software development using modern software engineering practices.
- **Design, develop and deploy production-ready AI applications** integrating FastAPI, OpenAI models, Retrieval-Augmented Generation (RAG), front-end web technologies, GitHub version control and cloud deployment.
- **Leverage generative AI throughout the software development lifecycle** to accelerate software development, data analysis, technical documentation and workflow optimisation while critically validating AI-generated outputs.
- **Design, develop and maintain professional websites and digital solutions** using GitHub Pages, WordPress (Elementor), HTML, CSS and JavaScript, integrating AI assistants, responsive design, modern deployment workflows and ongoing content management for academic and commercial clients.
- **Collaborate closely with researchers, industry clients and multidisciplinary teams** to deliver solutions aligned with project objectives, timelines and stakeholder expectations.

Honorary Senior Fellow

2026-2028 | [Faculty of Science](#), University of Melbourne, Parkville VIC, Australia

- **Contribute expertise in artificial intelligence, data science, bioinformatics and proteomics** through interdisciplinary research collaborations.

- Provide strategic advice on **analytical design, machine learning methodologies, statistical interpretation and research translation**.
- Support **publication development, grant applications**, mentoring and collaborative research initiatives involving complex biological datasets.
- **Develop and co-supervise Master's research projects in bioinformatics**, guiding project design, analytical methodology, computational workflows and scientific interpretation.

Senior Research Scientist

2013-2023 | [Agriculture Victoria Research](#) (AVR), Victorian Government, AgriBio Bundoora VIC, Australia

- Led large-scale multidisciplinary research projects involving **high-dimensional biological datasets, statistical modelling and computational analysis**.
- Designed innovative analytical strategies integrating proteomics, metabolomics, transcriptomics and bioinformatics to answer complex biological questions and support evidence-based decision making.
- Developed numerous **computational workflows and analytical pipelines** to improve data quality, reproducibility, automation and scientific interpretation.
- Managed multiple concurrent research projects, collaborating with **internal and external stakeholders** across universities, government agencies and international research organisations.
- **Supervised staff and students, managed Science A personnel**, delivered training workshops and mentored early-career researchers.
- Published extensively in high-impact journals while securing **external collaborations, competitive funding and international recognition**.

Post-doctoral Research Appointments

2009-2011 | [Australian National University](#) (ANU), Canberra ACT 0200, Australia

2007-2009 | [UMR BIOGECO INRA](#) 69 route d'Arcachon, 33612 Cestas, France

2005-2006 | [UMR Genetique Vegetale INRA](#), Ferme du Moulon, 91190 Gif-sur-Yvette, France

2003-2005 | [University of Nevada, Reno](#) (UNR), 1664 North Virginia Street, Reno NV 89557 USA

- Conducted internationally collaborative research in plant molecular biology, post-genomics and bioinformatics, establishing **expert analytical, computational and experimental foundations**.
- **Designed and executed innovative experimental studies**, applying advanced statistical analysis and computational approaches to address complex biological questions.
- Published peer-reviewed research in leading international journals while building an **extensive international collaboration network** across Australia, Europe and the United States.

EDUCATION

PhD, Plant Science (Proteomics)

2000-2003 | [Universite Paris-Saclay](#), Paris, France

- Thesis: *Proteomic analysis of response to water stress in maize based on cell differentiation*.
- **PhD doctorate** awarded with **Highest Distinctions**.
- Research combined experimental biology, quantitative proteomics, statistical analysis and computational interpretation of complex biological systems.

Honours, Plant Science (Genetic Resources and Crop Adaptation)

- 1998-1999 | [AgroParisTech](#), Paris, France

Masters, Plant Science (Biodiversity, Ecology, and Evolution)

- 1997-1998 | [Faculté des Sciences](#), Montpellier, France

Bachelor of Science (Life and Earth Sciences)

- 1994-1997 | [Faculté des Sciences](#), Montpellier, France

TECHNICAL SKILLS

Artificial Intelligence, Generative AI & LLM Engineering

- Practical experience designing and developing AI-powered applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI agents and modern AI engineering practices.
- Experience integrating OpenAI models into intelligent applications, developing AI-assisted analytical workflows, building conversational assistants, prompt engineering, structured prompting, grounding responses using external knowledge sources, evaluation of AI-generated outputs, and iterative optimisation of AI systems.
- Practical experience using both cloud-hosted and locally deployed language models (LM Studio and open-weight LLMs) for confidential AI workflows requiring secure offline processing, reproducibility and responsible AI practices.
- Ongoing professional development in enterprise AI architectures, cloud deployment and agentic AI technologies.

Core technologies OpenAI • ChatGPT • LM Studio • FastAPI • RAG • Hugging Face Transformers • GitHub Copilot • Gemini • Python • Prompt Design

Python, Software Development & Workflow Engineering

- Advanced Python developer with extensive experience building reproducible analytical software, automation pipelines and data-driven applications using Python, JupyterLab, VS Code, Git and GitHub.
- Experience includes API integration, FastAPI, workflow automation, web scraping, image processing, scientific computing, object-oriented programming and reproducible software development.
- Strong experience using pandas, NumPy, SciPy, scikit-learn, statsmodels, TensorFlow, PyTorch, XGBoost, LightGBM, Keras, matplotlib, BeautifulSoup, Selenium, Hugging Face Transformers and Pydantic.
- Additional experience with SQL, Bash, R and version-controlled collaborative software development.

Core technologies Python • JupyterLab • VS Code • Git • GitHub • FastAPI • SQL • Bash • Selenium • BeautifulSoup • Pydantic • pandas • NumPy • SciPy

Data Science, Machine Learning & Statistical Modelling

- Extensive experience transforming complex structured and unstructured datasets into actionable insights through statistical modelling, machine learning and predictive analytics.
- Expertise includes data cleaning, feature engineering, exploratory data analysis, regression, classification, clustering, dimensionality reduction, ensemble learning, deep learning and model validation.
- Strong analytical background spanning ANOVA, linear models, PCA, PLS, LDA, Random Forest, Support Vector Machines, XGBoost, neural networks, deep learning and advanced statistical inference.
- Experienced in selecting analytical methodologies according to individual project objectives rather than applying generic modelling approaches.

Core technologies scikit-learn • TensorFlow • PyTorch • XGBoost • LightGBM • Keras • statsmodels • Random Forest • Support Vector Machines • PCA • PLS • LDA • UMAP

Natural Language Processing & Text Analytics

- Experience developing NLP workflows for information extraction, semantic analysis and knowledge discovery using tokenisation, TF-IDF, word embeddings, transformer models and AI-assisted text analysis.
- Practical experience integrating NLP into scientific research, technical documentation, literature analysis and intelligent information retrieval workflows.

Core technologies Hugging Face Transformers • spaCy • TF-IDF • Word Embeddings • Tokenisation • Topic Modelling • Semantic Analysis • Prompt Engineering • Text Mining

Cloud Technologies, Deployment & DevOps

- Currently expanding enterprise cloud capabilities through AWS Educate with practical experience using Amazon S3, EC2, IAM, VPC, Lambda, RDS and SageMaker Studio Lab.
- Experience deploying and maintaining AI-enabled web applications using GitHub, GitHub Pages, Render and modern deployment workflows.

- Familiar with cloud-based AI development, version control, CI/CD concepts and deployment practices.

Core technologies AWS • SageMaker Studio Lab • Amazon S3 • Amazon EC2 • IAM • Amazon VPC • Amazon RDS • AWS Lambda • GitHub Pages • Render

Data Visualisation & Decision Support

- Experienced in transforming complex analytical outputs into intuitive visual stories for technical and non-technical audiences through dashboards, publication-quality graphics and interactive reporting.
- Tools include Python, R, Tableau and Power BI. Expertise includes exploratory visual analytics, executive reporting, workflow diagrams and interactive dashboards supporting evidence-based decision making.

Core technologies Tableau • Power BI • matplotlib • Plotly • seaborn • pandas • Excel • JupyterLab

Web Development & Digital Solutions

- Design, develop and maintain professional websites and AI-enabled digital platforms using HTML, CSS, JavaScript, PHP, GitHub Pages, WordPress (Elementor), cPanel and WebCentral hosting environments.
- Experience integrating AI assistants, chatbot technologies, responsive design, secure email handling, hosting administration and ongoing website optimisation to improve client engagement and digital communication.

Core technologies HTML • CSS • JavaScript • PHP • WordPress • Elementor • GitHub Pages • cPanel • WebCentral

Bioinformatics & Scientific Data Science

- Specialist expertise applying AI, machine learning and advanced computational methods to large-scale biological datasets including proteomics, metabolomics, transcriptomics and multi-omics integration.
- Extensive experience in biological data mining, genome/protein mapping, functional annotation, pathway analysis and scientific interpretation using recognised bioinformatics platforms and databases.

Core technologies BLAST+ • Cytoscape • AlphaFold • Apollo/JBrowse • Galaxy • UniProtKB • Gene Ontology • STRING • KEGG • Reactome • MetaboAnalyst • ProteoWizard

PROFESSIONAL PORTFOLIO

My professional portfolio demonstrates the practical application of artificial intelligence, data science, machine learning and scientific software development through:

- Professional [website](#) showcasing custom-built AI applications, including a Retrieval-Augmented Generation (RAG) chatbot that answers visitor questions using website content, publications and professional expertise.
- Active [GitHub](#) profile containing 37 repositories covering Python, R, SQL, web development, machine learning and reproducible analytical workflows.
- [Google Scholar](#) profile documenting 71 scientific publications with over 3,500 citations (h-index 22).
- [Tableau Public](#) dashboards demonstrating interactive data visualisation and analytical storytelling.
- 13 publicly available research datasets and community resources supporting reproducible scientific research and open science initiatives via Zenodo, (Giga)ⁿ DB, PRIDE, and MassIVE repositories.
- Created publicly accessible proteogenomics visualisation resources ([Apollo/JBrowse](#)).

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

Amazon Web Services (AWS) Cloud & Artificial Intelligence

AWS Educate (2026–Present)

Continuously expanding enterprise cloud and AI capabilities through structured AWS Educate learning pathways.

Completed the full Getting Started learning series covering:

- AWS Management Console
- Cloud Computing Foundations
- Storage (Amazon S3)
- Compute (Amazon EC2)
- Networking (Amazon VPC)
- Databases (Amazon RDS)
- Security (IAM)

- Cloud Operations
- Serverless Computing (AWS Lambda)

Current professional development focuses on:

- Artificial Intelligence & Machine Learning
- AWS Core Concepts
- SageMaker Studio Lab
- AWS Free Tier practical environments
- Cloud deployment and AI infrastructure
- Preparing for AWS Certified AI Practitioner (AIF-C01)

Artificial Intelligence & Data Science

CodeCademy Professional Certifications (2024–Present)

Completed professional training in:

- Career Path: Data Scientist – Analytics Specialist
- Career Path: Data Scientist – Machine Learning Specialist (Python)
- Skill Path: Build Chatbots with Python
- Skill Path: Build Websites with HTML, CSS & GitHub Pages
- Introduction to Prompt Engineering
- Introduction to Large Language Models (LLMs)

Current professional development focuses on:

- Career Path: Data Scientists: Natural Language Processing (NLP)

Leadership, Mentoring & Training

Throughout my career:

- Managed Science staff and supervised students in Australia, the USA and France.
- Mentored early-career researchers and STEM participants.
- Delivered training workshops in Systems Biology and computational analysis.
- Guest lecturer and demonstrator in Australia and France.
- Mentor for National Young Indigenous Women's STEM Academy.

Scientific Leadership & Professional Service

Demonstrates sustained leadership within the scientific community through editorial and committee activities.

Editorial Leadership (2021–Present)

Editor for Research Topics and Special Issues covering proteomics, plant science and systems biology for the following scientific journals:

- BMC Plant Biology
- Frontiers in Plant Science
- Proteomes
- Biomolecules
- International Journal of Molecular Sciences

Scientific Committees (2019-2023)

Board member for:

- Proteomics & Metabolomics Victoria (PMV)
- EU COST Action — Protein Production in Photosynthetic Organisms

Committed to continuous professional development across AI, data science, machine learning and bioinformatics, translating emerging technologies into practical, responsible and reproducible solutions through scientific rigour, modern software engineering and collaborative innovation.
